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Assembly and computer simulation by Geant4 of a ^{3}He detector for cosmic radiation in the ground

Ranulfo da Silva Dias[1,2], Thiago Augusto do Carmo[1,2], Felipe Lopes Frigi[1,2], Guilherme Gazolla Santana[1,2], Roberto Bedogni[5], Odair Lélis Gonçalez[1,2], Maurício Tizziani Pazianotto[1,2], Edson Ramos de Andrade[2,3,4], Claudio Antonio Federico[1,2]

[1]Aeronautics Technological Institute (ITA), Brazilian Air Force, São Paulo, Brazil
[2]Institute for Advanced Studies (IEAv), Brazilian Air Force, Sao Paulo, Brazil
[3]Defense Engineering Graduate Program, Military Institute of Engineering (IME), Rio de Janeiro, Brazil
[4]University of Western Sao Paulo, (UNOESTE), Sao Paulo, Brazil
[5]National Institute of Nuclear Physics - Frascati National Laboratories (LNF), Rome, Italy

***Correspondence to: claudio.federico40@gmail.com**

1. INTRODUCTION

- **Primary cosmic radiation**: mesons, muons, electrons, photons, and **neutrons**.

- **Neutrons** are the most relevant for monitoring:
- High penetrating power.
- Significant contribution to the **ambient dose equivalent** at high altitudes and ground level.

- Proposal: **Neutron detection system** with **^{3}He proportional tubes**
- **Experimental tests** at LRI/IEAv using a **^{241}Am-Be source** to verify response.
- **Monte Carlo simulations (Geant4)** developed
- **Final installation**: Container at **Pico dos Dias Observatory** (Brazópolis-MG, 1840 m altitude).
- Continuous monitoring of **cosmic radiation in Brazil**, supporting:
- **Aerospace applications**.
- Studying specific regions, such as the **South Atlantic Magnetic Anomaly (SAMA)**, where the Earth's magnetic field differs from the rest of the globe.
- **Radiological safety studies** in commercial flights.

2. MATERIALS AND METHODS

The neutron detector system consists of two polyethylene moderator layers, one upper and one lower (73.9 × 75.5 × 9.5 cm), with a central lead layer (73.9 × 74.5 × 3 cm), whose function is to promote the multiplication of high-energy neutrons.

The base is an iron plate (95.2 × 95.3 × 1.0 cm), supported on a wooden pallet containing slats arranged in three bases, providing stable support. Around the base, there is a hollow polyethylene box, similar to a wall (89.2 × 89.2 × 13 cm; thickness de 4.1 cm) covered by a hollow lead box (80 × 80 × 12.5 cm; thickness 3 cm).

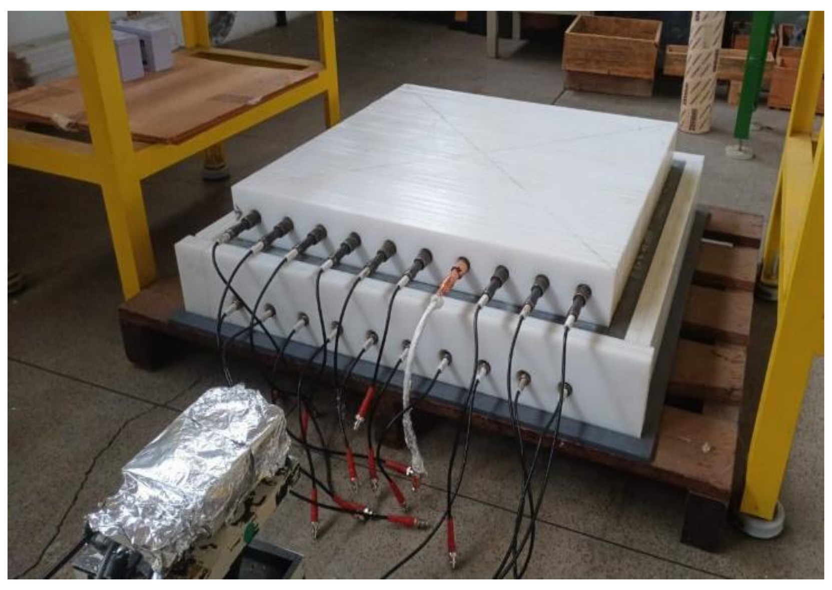

Figure 1: View of the system in the laboratory

The detectors used were manufactured by **Harshaw (Allied Chemical Corporation) in 1975** and are 71.8 cm long and 25.55 mm in diameter. The connectors are 4.2 cm long and 1.0 cm in diameter. The active region filled with ^{3}He gas has an approximate length between 64 and 66 cm.

The **simulation was performed via Monte Carlo using Geant4** software (version 11.02).

- **Source alignment**: positioned on the central axis of the upper polyethylene plate, 68 cm from detector center.
- **Emission setup**: isotropic emission within a **55° cone angle** (covering full detector area) and **75° cone angle** (back and right side). In **Geant4**, 55° and 75° are the **half-angles** between the cone axis and its wall. **Correction factor** applied for the **solid angle fraction** relative to **4π steradians**.
- **Simulation statistics**: **2.000.000.000 events** simulated.

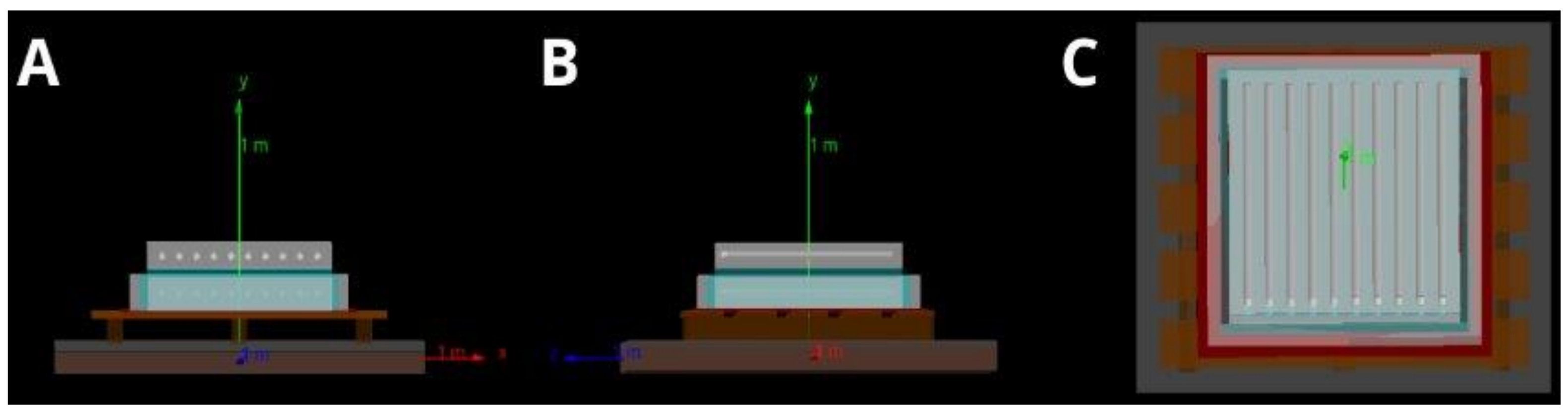

Figure 2: Visualization in Geant4: A) Front; B) Side; C) Top

3. RESULTS

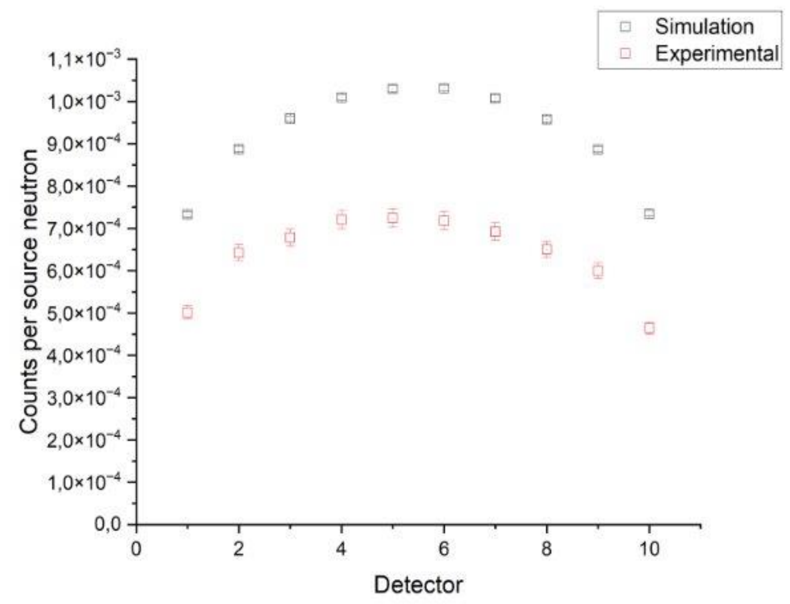

Figure 3: Central incidence for the 4 atm

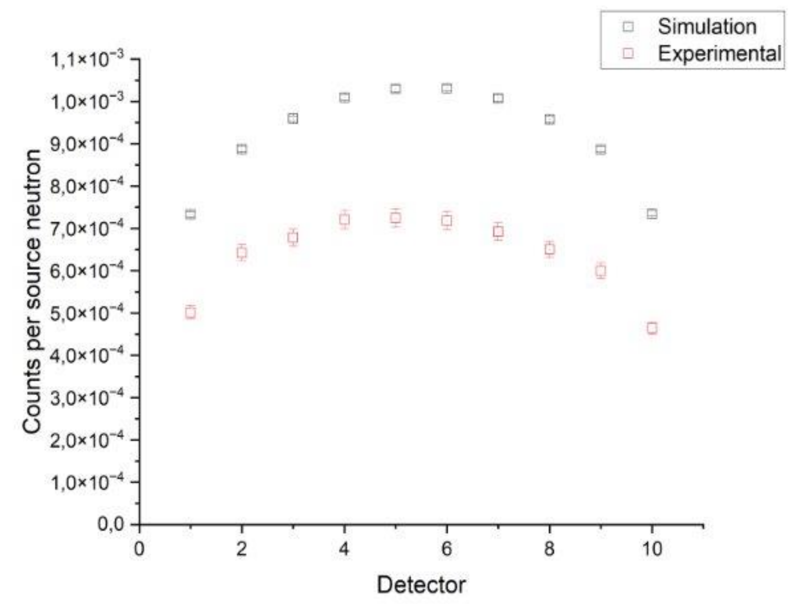

Figure 4: Central incidence for the 6 atm

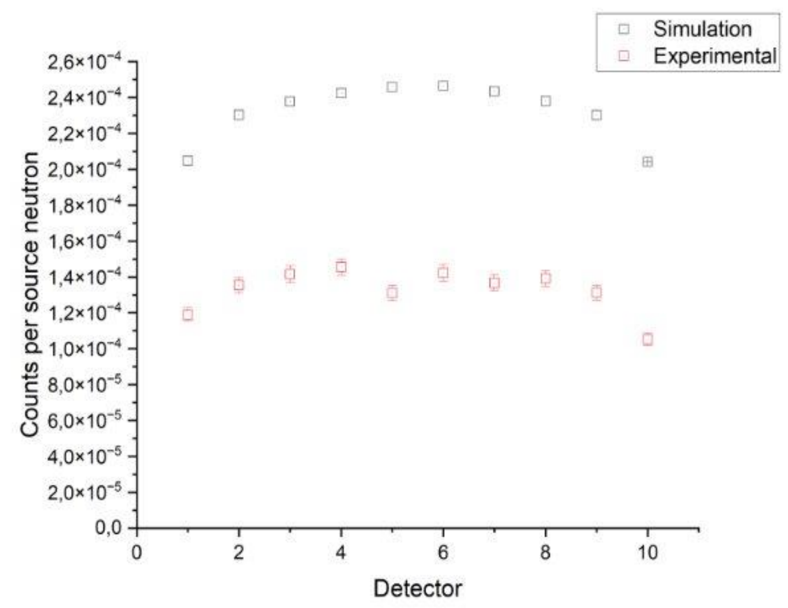

Figure 5: Side incidence for the 4 atm

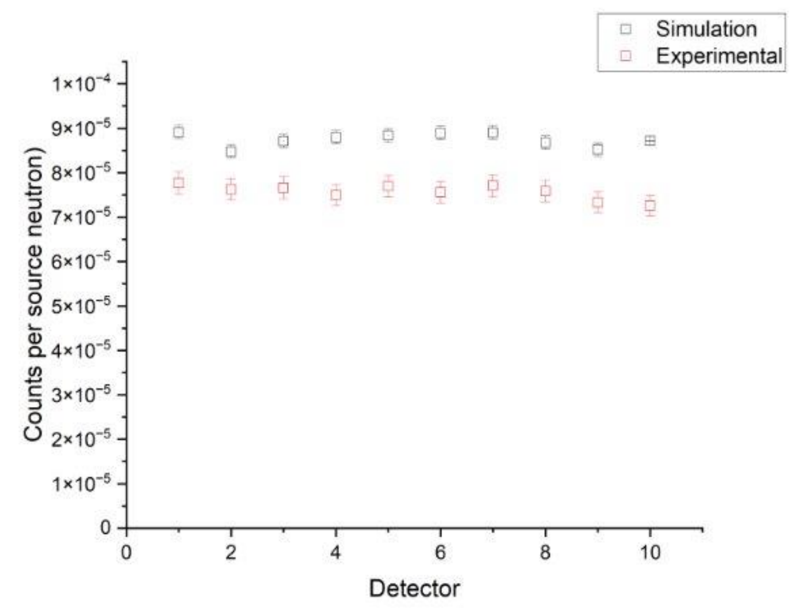

Figure 6: Side incidence for the 6 atm

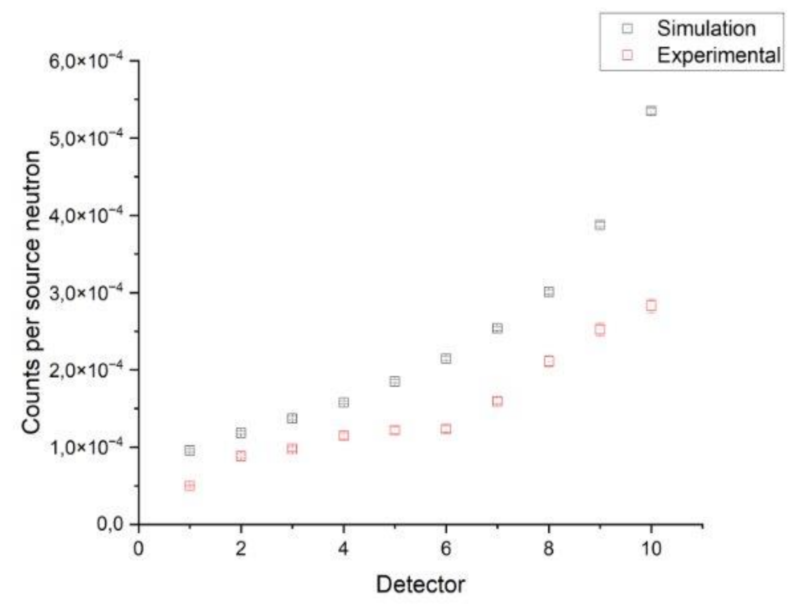

Figure 7: Lateral incidence for the 4 atm

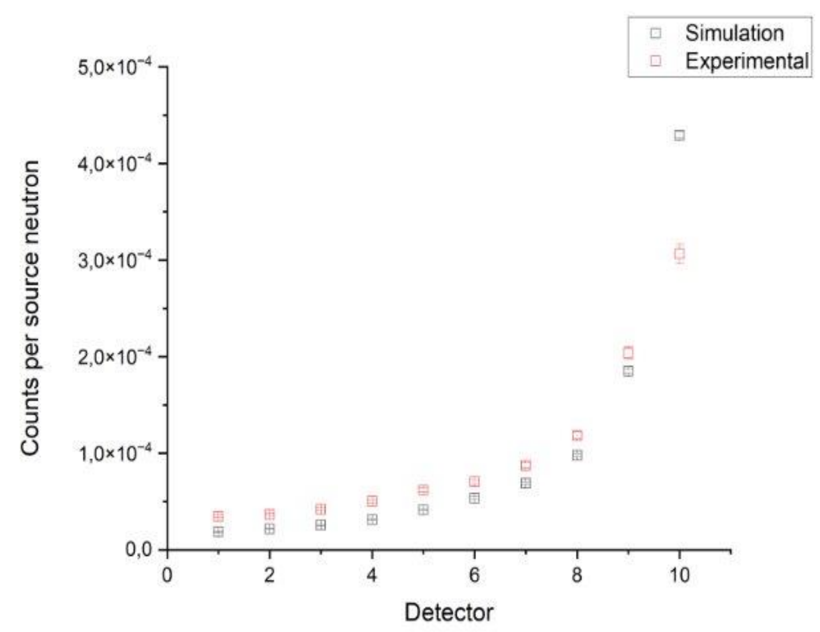

Figure 8: Lateral incidence for the 6 atm

4. CONCLUSIONS

Experimental tests showed good uniformity among detectors (variation of 3–5%). Geant4 simulations reproduced experimental behavior, though with discrepancies still under investigation. The model proved reliable and will be refined, especially regarding material density. Next steps: model adjustments, detailing of internal components, and estimation of the system's response to the cosmic radiation spectrum at the operating site.

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